$X' = AX$ $\quad n = 2$ $\quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

1) $\lambda_1, \lambda_2 \in \mathbb{R}$ $\lambda_1 \neq \lambda_2$
2) $\lambda_1, \lambda_2 \in \mathbb{R}$ $\lambda_1 = \lambda_2$
3) $\lambda_1, \lambda_2 \in \mathbb{C} - \mathbb{R}$, $\bar{\lambda}_1 = \lambda_2$

2) $\sigma(A) = \{\lambda\}$ $\quad \dim_{\mathbb{R}} V_\lambda = 1$

$v_2 \in \mathbb{R}^2$, $v_2 \notin V_\lambda \Rightarrow v_1 = (A - \lambda I_2) v_2 \in V_\lambda$

$X_1(t) = e^{\lambda t} v_1$, $\quad X_2(t) = e^{\lambda t}(t v_1 + v_2)$ sistem fundamental de soluții pt $X' = AX$ (bază în S_0)

$X(t) = C_1 X_1(t) + C_2 X_2(t)$, $C_1, C_2 \in \mathbb{R}$ sol. gen. a $X' = AX$

exemplu 1) $A = \begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix}$, $\lambda = 3$ $\quad \operatorname{tr} A = 6$, $\det A = 9$

$$\begin{cases} x' = 2x - y \\ y' = x + 4y \end{cases} \quad \begin{cases} x(0) = 0 \\ y(0) = 0 \end{cases} \Rightarrow \begin{cases} x(t) = 0 \\ y(t) = 0 \end{cases} \quad \forall t \in \mathbb{R}$$

$$\begin{cases} x(0) = 1 \\ y(0) = 0 \end{cases}$$

$\lambda = 3 \qquad A = \begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix} \qquad V_1 \qquad \begin{cases} 2x - y = 3x \\ x + 4y = 3y \end{cases} \Rightarrow y = -x$

$$V_1 = \left\{ \begin{pmatrix} x \\ -x \end{pmatrix} / x \in \mathbb{R} \right\}$$

Je me $v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ $\qquad v_1 = (A - \lambda I_2) v_2 = \left(\begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \right) \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$\left(v_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$

$$= \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \end{pmatrix} \in V_1$$

$$\Rightarrow \quad X_1(t) = e^{\lambda t} v_1 = e^{3t} \begin{pmatrix} -3 \\ 3 \end{pmatrix}$$

$$X_2(t) = e^{\lambda t} (t v_1 + v_2) = e^{3t} \left(t \begin{pmatrix} -3 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right)$$

$$\Rightarrow \quad X(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 e^{3t} \begin{pmatrix} -3 \\ 3 \end{pmatrix} + C_2 e^{3t} \left(t \begin{pmatrix} -3 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right)$$

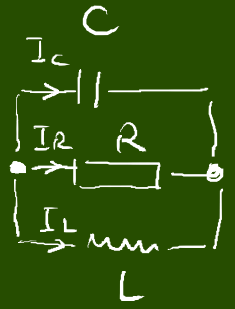
$$\begin{cases} x(t) = -3 C_1 e^{3t} + C_2 e^{3t} (-3t + 1) \\ y(t) = 3 C_1 e^{3t} + C_2 e^{3t} (3t + 2) \end{cases}$$

$x(0) = 1$
$y(0) = 0$
$\Rightarrow$
$$\begin{cases} -3C_1 + C_2 = 1 \\ 3C_1 + 2C_2 = 0 \end{cases}$$

$5C_2 = 1, \quad C_2 = 1/5$

$C_1 = -\frac{2}{3}C_2 = -\frac{2}{15}$

exemple 2)



$I_C + I_R + I_L = 0$.

$U_C = U_R = U_L$

$$\begin{cases} I_C(t) = C \cdot Q(t) = C \cdot U_C'(t) \\ I_R(t) = \dfrac{U_R(t)}{R} \\ U_L(t) = L \cdot I_L'(t) \end{cases}$$

$I_L'(t) = \frac{1}{L} U_L(t)$

$U_L'(t) = U_C'(t) = \dfrac{I_C(t)}{C} = \dfrac{(-I_R(t) - I_L(t))}{C} = -\dfrac{I_L(t)}{C} - \dfrac{U_L(t)}{RC}$

$x = I_L \qquad y = U_L$

$$\begin{cases} x' = \frac{1}{L} y \\ y' = -\frac{1}{C} x - \frac{1}{RC} y \end{cases}$$

$R = 1\,\Omega$, $C = 1\,F$, $L = 4\,H$

$$\begin{cases} x' = \frac{1}{4} y \\ y' = -x - y \end{cases}$$

$x(0) = 1A \qquad y(0) = 2V$

$$A = \begin{pmatrix} 0 & 1/4 \\ -1 & -1 \end{pmatrix} \qquad P_A(x) = x^2 + x + \frac{1}{4} = \left(x + \frac{1}{2}\right)^2$$

$\lambda = -\frac{1}{2} \in \mathbb{R}$ $\qquad V_\lambda$:

$$\begin{cases} \frac{1}{4} y = -\frac{1}{2} x \\ -x - y = -\frac{1}{2} y \end{cases} \Rightarrow y = -2x$$

$$V_\lambda = \left\{ \begin{pmatrix} x \\ -2x \end{pmatrix} \mid x \in \mathbb{R} \right\}$$

$v_2 \in \mathbb{R}^2 \qquad v_2 \notin V_\lambda \qquad v_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$

$$v_1 = (A - \lambda I_2) v_2 = \begin{pmatrix} 1/2 & 1/4 \\ -1 & -1/2 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$X_1(t) = e^{\lambda t} v_1 = e^{-t/2}\begin{pmatrix}1\\-2\end{pmatrix}$$

$$X_2(t) = e^{\lambda t}(tv_1 + v_2) = e^{-t/2}\left(t\begin{pmatrix}1\\-2\end{pmatrix} + \begin{pmatrix}2\\0\end{pmatrix}\right)$$

$$\begin{pmatrix}x(t)\\y(t)\end{pmatrix} = c_1 e^{-t/2}\begin{pmatrix}1\\-2\end{pmatrix} + c_2 e^{-t/2}\left(t\begin{pmatrix}1\\-2\end{pmatrix} + \begin{pmatrix}2\\0\end{pmatrix}\right) \xrightarrow{t\to+\infty} \begin{pmatrix}0\\0\end{pmatrix} \quad c_1, c_2 \in \mathbb{R}$$

$$\begin{cases} x(0) = 1 \\ y(0) = 2 \end{cases} \qquad \begin{cases} c_1 + 2c_2 = 1 \\ -2c_1 = 2 \end{cases} \qquad \begin{matrix} c_2 = 1 \\ c_1 = -1 \end{matrix}$$

$$x(t) = -e^{-t/2} + e^{-t/2}(t+2) = e^{-t/2}(t+1) \searrow 0$$

$$y(t) = 2e^{-t/2} + e^{-t/2}(-2t) = e^{-t/2}(-2t+2) \longrightarrow 0$$

Caso 3 Valori propri complessi $\lambda = \alpha + i\beta$ $\bar{\lambda} = \alpha - i\beta$ $\beta \neq 0$

$a(\lambda) = a(\bar{\lambda}) = 1 \Rightarrow$ A diagonalizzabile per $\mathbb{C}$

$\dim_{\mathbb{C}} V_\lambda = 1$ $V_\lambda = \mathrm{Sp}\{v\} \Rightarrow V_{\bar{\lambda}} = \mathrm{Sp}\{\bar{v}\}$

$Z(t) = e^{\lambda t} v$ solução (complexa) a sist. de ec. dif $X' = AX$

$X_1(t) = \operatorname{Re} Z(t)$ $\quad$ $X_2(t) = \operatorname{Im} Z(t)$ $\quad$ fcs. reais $\quad \mathbb{R} \longrightarrow \mathbb{R}^2$

$Z' = AZ \qquad Z = X_1 + iX_2$

$X_1' + iX_2' = A(X_1 + iX_2) \Rightarrow X_1' = AX_1$

$X_2' = AX_2$

$X_1, X_2 \in S_0$

X_1, X_2 lin. indep. $\iff$ $X_1(0), X_2(0) \in \mathbb{R}^2$ lin. indep.

$X_1(0) = v_1 \qquad X_2(0) = v_2$

$v = v_1 + iv_2 \in V_\lambda$

$\bar{v} = v_1 - iv_2 \in V_{\bar{\lambda}}$

Se $\alpha_1, \alpha_2 \in \mathbb{R}$ a.i. $\alpha_1 v_1 + \alpha_2 v_2 = 0$

$$\alpha_1 \cdot \frac{v + \bar{v}}{2} + \alpha_2 \cdot \frac{v - \bar{v}}{2i} = 0$$

$$\underbrace{\left(\frac{\alpha_1}{2} + \frac{\alpha_2}{2i}\right)}_{0} v + \underbrace{\left(\frac{\alpha_1}{2} - \frac{\alpha_2}{2i}\right)}_{0} \bar{v} = 0 \quad \Big| \Rightarrow$$

$v, \bar{v}$ liniar independente

$\Rightarrow \frac{\alpha_1}{2} + \frac{\alpha_2}{2i} = 0, \quad \frac{\alpha_1}{2} - \frac{\alpha_2}{2i} = 0 \Rightarrow \alpha_1 = \alpha_2 = 0$

$\Rightarrow v_1, v_2$ liniar independente $\Rightarrow X_1, X_2$ liniar independente.

$\Rightarrow$ $X_1(t), X_2(t)$ Sistem fundamental de soluții pt $X' = AX$

Sol. gen. $X(t) = C_1 X_1(t) + C_2 X_2(t)$

$Z(t) = e^{\lambda t} v = e^{(\alpha + i\beta)t}(v_1 + iv_2) = e^{\alpha t}(\cos\beta t + i \sin\beta t)(v_1 + iv_2)$

$\lambda = \alpha + i\beta \quad v = v_1 + iv_2$

$e^{it} = \cos t + i \sin t$ (Euler)

$$\Rightarrow \begin{cases} X_1(t) = e^{\alpha t}(\cos\beta t\, v_1 - \sin\beta t\, v_2) \\ X_2(t) = e^{\alpha t}(\sin\beta t\, v_1 + \cos\beta t\, v_2) \end{cases}$$

exemplo ecossistema 2 espécies predador / presa

$x(t)$, $y(t)$ t tempo (ano)

$$\begin{cases} x' = 0{,}2x - y \\ y' = x + 0{,}2y \end{cases} \qquad x(0) = 10 \qquad y(0) = 2$$

$$A = \begin{pmatrix} 0{,}2 & -1 \\ 1 & 0{,}2 \end{pmatrix} \qquad P_A(x) = X^2 - 0{,}4X + (0{,}04 + 1) = X^2 - 0{,}4X + 1{,}04$$

$$\Delta = 0{,}16 - 4{,}16 = -4$$

$$\lambda = \frac{0{,}4 \pm 2i}{2} = 0{,}2 \pm i$$

$\lambda = \underset{\alpha}{0{,}2} + i \underset{\beta}{1}$ $\qquad V_\lambda : \begin{cases} 0{,}2x - y = (0{,}2 + i)x \\ x + 0{,}2y = (0{,}2 + i)y \end{cases} \Rightarrow y = -ix$

$$V_\lambda = \left\{ \begin{pmatrix} x \\ -ix \end{pmatrix} \mid x \in \mathbb{C} \right\} = Sp\left\{ \underbrace{\begin{pmatrix} 1 \\ -i \end{pmatrix}}_{v} \right\}$$

$$v = \begin{pmatrix} 1 \\ -i \end{pmatrix} = \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{v_1} + i \underbrace{\begin{pmatrix} 0 \\ -1 \end{pmatrix}}_{v_2}$$

$\Rightarrow \quad X_1(t) = \mathrm{Re}\left(e^{\lambda t} v \right) = \mathrm{Re}\left(e^{(0,2+i)t} \begin{pmatrix} 1 \\ -i \end{pmatrix} \right)$

$X_2(t) = \mathrm{Im}\left(\quad \right)$

$e^{(0,2+i)t} \begin{pmatrix} 1 \\ -i \end{pmatrix} = e^{0,2t} (\cos t + i \sin t) \begin{pmatrix} 1 \\ -i \end{pmatrix} = e^{0,2t} \begin{pmatrix} \cos t + i \sin t \\ \sin t - i \cos t \end{pmatrix}$

$X_1(t) = e^{0,2t} \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} \qquad X_2(t) = e^{0,2t} \begin{pmatrix} \sin t \\ -\cos t \end{pmatrix}$

$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = X(t) = C_1 X_1(t) + C_2 X_2(t) = \left(C_1 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + C_2 \begin{pmatrix} \sin t \\ -\cos t \end{pmatrix} \right) e^{0,2t}$

$x(0) = 10, \quad y(0) = 2 \quad \Rightarrow \begin{cases} C_1 = 10 \\ -C_2 = 2 \end{cases}$

$\begin{cases} x(t) = (10 \cos t - 2 \sin t) e^{0,2t} \\ y(t) = (10 \sin t + 2 \cos t) e^{0,2t} \end{cases} \qquad t \geq 0$

$x(t) = 0 \Leftrightarrow \tan t = 5$

$t \geq \frac{\pi}{3} \approx 1,..$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \left(C_1 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + C_2 \begin{pmatrix} \sin t \\ -\cos t \end{pmatrix} \right) e^{0.2t}$$

$$= e^{0,2t} \underbrace{\begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix}}_{\text{matrice ortogonală}} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} \qquad C_1, C_2 \in \mathbb{R}$$

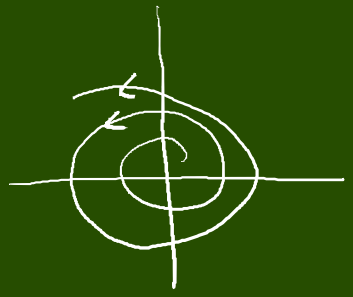
$$\left\| \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} \right\| = \sqrt{x^2 + y^2} = e^{0.2t} \left\| \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} \right\| = e^{0.2t} \sqrt{C_1^2 + C_2^2}$$

$$\Rightarrow \quad x^2 + y^2 = e^{0.2t} \underbrace{(C_1^2 + C_2^2)}_{R^2}$$

$e^{0.2t} \to \infty$



Spirală logaritmică care se îndepărtează de origine



sau



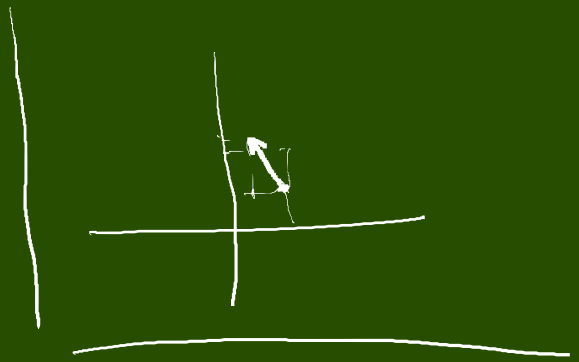
?

$$\begin{cases} x' = 0,2x - y \\ y' = x + 0,2y \end{cases}$$

$x = 1, y = 1$

$$\begin{cases} x' = -0,8 \\ y' = 1,2 \end{cases}$$

Système de éq. dif. linéaires à coefficients constants non homogènes



$$X' = AX + b(t)$$

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \qquad b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} : I \longrightarrow \mathbb{R}^n \text{ cont.}$$

Thm. Pb Cauchy $\begin{cases} X' = AX + b(t) \\ X(t_0) = X_0 \end{cases}$ a sol. unique $\forall\, t_0 \in I,\ \forall X_0 \in \mathbb{R}^n$

Prop. $S_0 = \{X : \mathbb{R} \longrightarrow \mathbb{R}^n \mid X \text{ dif},\ X' = AX\}$ sp. vect. $\dim_{\mathbb{R}} S_0 = n$

$S_{n0} = \{X : I \longrightarrow \mathbb{R}^n \mid X \text{ dif},\ X' = AX + b(t)\}$ multiple.

Fie $X_p \in S_{n0}$ (sol. particulară)

Atunci o funcție X satisface $X \in S_{n0} \iff X - X_p \in S_0$

$$X' = AX + b(t)$$
$$X_p' = AX_p + b(t)$$
$$(X - X_p)' = A(X - X_p)$$

Consec. Fie $X_p \in S_{an}$ o sol. particulară $\Rightarrow$ sol. generală a sist. neomogen $X' = AX + b(t)$ este de forma

$$X(t) = X_p(t) + X_0(t)$$ unde $X_0(t)$ este sol. generală a sist. omogen $X' = AX$, $X_0(t) = C_1 X_1(t) + \dots + C_n X_n(t)$

$\{X_1, \dots X_n\}$ bază pt S_0.

Metoda variației constantelor pentru determinarea soluției sistemului de ec. dif. liniare neomogene:

$$X' = AX + b(t)$$

1) $X' = AX \longrightarrow X_1(t), \dots, X_n(t)$ baza pt sp. sol. S_0

$\longrightarrow X_0(t) = C_1 X_1(t) + \dots + C_n X_n(t)$, $C_1 \dots C_n \in \mathbb{R}$

2) caut sol. sist. neomogen de forma $X(t) = C_1(t)X_1(t) + \dots + C_n(t)X_n(t)$

$C_1'(t)X_1(t) + C_1(t)X_1'(t) + \dots + C_n'(t)X_n(t) + C_n(t)X_n'(t) =$

$= A(C_1(t)X_1(t) + \dots + C_n(t)X_n(t)) + b(t)$

$\Rightarrow C_1'(t)X_1(t) + \dots + C_n'(t)X_n(t) = b(t)$ $\quad (X_1(t)/\dots/X_n(t))$ matrice inversabilă

$\Rightarrow C_1'(t) = \dots \quad \dots \quad C_n'(t) = \dots$

$C_1(t) = \dots \quad C_n(t) = \dots$

exemplu.

$$\begin{cases} x' = x + 2y + 3 \\ y' = 2x + y \end{cases} \qquad \begin{matrix} x(0) = 3 \\ y(0) = 2 \end{matrix}$$

$$\begin{cases} x' = x + 2y \\ y' = 2x + y \end{cases} \qquad A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \qquad \begin{aligned} \lambda_1 + \lambda_2 &= 2 \\ \lambda_1 \lambda_2 &= -3 \end{aligned} \quad \Rightarrow \quad \begin{aligned} \lambda_1 &= 3 \\ \lambda_2 &= -1 \end{aligned}$$

$\lambda_1 = 3$:

$$\begin{cases} x + 2y = 3x \\ 2x + y = 3y \end{cases} \Rightarrow x = y \qquad V_{\lambda_1} = Sp\left\{ \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{v_1} \right\}$$

$\lambda_2 = -1$:

$$V_{\lambda_2} = Sp\left\{ \underbrace{\begin{pmatrix} 1 \\ -1 \end{pmatrix}}_{v_2} \right\}$$

$$\Rightarrow \quad X_1(t) = e^{\lambda_1 t} v_1 = e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad X_2(t) = e^{\lambda_2 t} v_2 = e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$X_0(t) = \begin{pmatrix} x_0(t) \\ y_0(t) \end{pmatrix} = C_1 X_1(t) + C_2 X_2(t)$$

$$= C_1 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$x_0(t) = C_1 e^{3t} + C_2 e^{-t}$$

$$y_0(t) = C_1 e^{3t} - C_2 e^{-t}$$

$$\begin{cases} x' = x + 2y + 3 \\ y' = 2x + y \end{cases} \qquad \begin{aligned} x(t) &= C_1(t)e^{3t} + C_2(t)e^{-t} \\ y(t) &= C_1(t)e^{3t} - C_2(t)e^{-t} \end{aligned}$$

$$\Rightarrow \begin{cases} C_1'(t)e^{3t} + C_2'(t)e^{-t} = 3 \\ C_1'(t)e^{3t} - C_2'(t)e^{-t} = 0 \end{cases}$$

$$2C_1'(t)e^{3t} = 3 \qquad C_1'(t) = \frac{3}{2}e^{-3t} \Rightarrow C_1(t) = -\frac{1}{2}e^{-3t}$$

$$2C_2'(t)e^{-t} = 3 \qquad C_2'(t) = \frac{3}{2}e^{t} \Rightarrow C_2(t) = \frac{3}{2}e^{t}$$

$$\Rightarrow \begin{cases} x_p(t) = -\frac{1}{2}e^{-3t}\cdot e^{3t} + \frac{3}{2}e^{t}\cdot e^{t} = 1 \\ y_p(t) = -\frac{1}{2}e^{-3t}\cdot e^{3t} - \frac{3}{2}e^{t}\cdot e^{t} = -2 \end{cases}$$

$$\Rightarrow \begin{cases} x(t) = x_0(t) + x_p(t) = C_1e^{3t} + C_2e^{-t} + 1 \\ y(t) = y_0(t) + y_p(t) = C_1e^{3t} - C_2e^{-t} - 2 \end{cases}$$

$x(0) = 3$, $y(0) = 2$ $\Rightarrow \begin{cases} C_1 + C_2 + 1 = 3 \\ C_1 - C_2 - 2 = 2 \end{cases}$

$2C_1 - 1 = 5 \quad C_1 = 3$

$C_2 = -1$

exemple. $\begin{cases} x' = -x + 4y \\ y' = -2x + 3y + 2e^t \end{cases}$

$x(0) = 0$
$y(0) = 0$ $\not\Rightarrow$!!! $x(t) = 0$ $\forall t$
$y(t) = 0$

$\begin{cases} x' = -x + 4y \\ y' = -2x + 3y \end{cases}$ $\quad A = \begin{pmatrix} -1 & 4 \\ -2 & 3 \end{pmatrix}$ $\quad P_A(X) = X^2 - 2X + 5 = (X-1)^2 + 4 \begin{cases} 1+2i \\ 1-2i \end{cases}$

$\lambda = 1 + 2i$ $\quad \begin{cases} -x + 4y = (1+2i)x \\ -2x + 3y = (1+2i)y \end{cases} \Rightarrow 4y = 2(1+i)x$

$y = \frac{1+i}{2} x$

$V_\lambda = Sp\left\{ \overbrace{\begin{pmatrix} 2 \\ 1+i \end{pmatrix}}^{v} \right\}$

X_1, X_2 $\qquad Z(t) = e^{\lambda t} v = e^{(1+2i)t} \begin{pmatrix} 2 \\ 1+i \end{pmatrix}$

$$= e^t (\cos 2t + i \sin 2t) \begin{pmatrix} 2 \\ 1+i \end{pmatrix}$$

$$= e^t \begin{pmatrix} 2\cos 2t + 2i \sin 2t \\ \cos 2t - \sin 2t + i(\cos 2t + \sin 2t) \end{pmatrix}$$

$$X_1(t) = e^t \begin{pmatrix} 2\cos 2t \\ \cos 2t - \sin 2t \end{pmatrix}$$

$$X_2(t) = e^t \begin{pmatrix} 2\sin 2t \\ \cos 2t + \sin 2t \end{pmatrix}$$

$$X_0(t) = \begin{pmatrix} x_0(t) \\ y_0(t) \end{pmatrix} = C_1 e^t \begin{pmatrix} 2\cos 2t \\ \cos 2t - \sin 2t \end{pmatrix} + C_2 e^t \begin{pmatrix} 2\sin 2t \\ \cos 2t + \sin 2t \end{pmatrix}$$

$$\begin{cases} x_0(t) = 2C_1 e^t \cos 2t + 2C_2 e^t \sin 2t \\ y_0(t) = C_1 e^t (\cos 2t - \sin 2t) + C_2 e^t (\cos 2t + \sin 2t) \end{cases}$$

$$\begin{cases} x_p(t) = 2C_1(t)e^t \cos 2t + 2C_2(t)e^t \sin 2t \\ y_p(t) = C_1(t)e^t(\cos 2t - \sin 2t) + C_2(t)e^t(\cos 2t + \sin 2t) \end{cases}$$

soluție a sist. neom.

$\Rightarrow$ $C_1(t), C_2(t)$ verifică sistemul:

$$\begin{cases} 2C_1'(t)e^t \cos 2t + 2C_2'(t)e^t \sin 2t = 0 \\ C_1'(t)e^t(\cos 2t - \sin 2t) + C_2'(t)e^t(\cos 2t + \sin 2t) = 2e^t \end{cases}$$

$$\begin{cases} C_1'(t)\cos 2t + C_2'(t)\sin 2t = 0 \quad (*) \\ C_1'(t)(\cos 2t - \sin 2t) + C_2'(t)(\cos 2t + \sin 2t) = 2 \end{cases}$$

$$C_1'(t)\sin 2t - C_2'(t)\cos 2t = -2 \quad (**)$$

$(*),(**) \Rightarrow$

$$\begin{cases} C_1'(t)\cos 2t + C_2'(t)\sin 2t = 0 \quad / \cos 2t, \sin 2t \\ C_1'(t)\sin 2t - C_2'(t)\cos 2t = -2 \quad / \sin 2t, \cos 2t \end{cases}$$

$$C_1'(t) = -2\sin 2t \Rightarrow C_1(t) = \cos 2t$$

$$C_2'(t) = 2\cos 2t \quad \Rightarrow C_2(t) = \sin 2t$$

$$\begin{cases} x_p(t) = 2\cos 2t \cdot e^t \cos 2t + 2\sin 2t\, e^t \sin 2t = 2e^t \\ y_p(t) = \cos 2t\, e^t(\cos 2t - \sin 2t) + \sin 2t\, e^t(\cos 2t + \sin 2t) = e^t \end{cases}$$

$\Rightarrow$ sol. generală $\begin{cases} x(t) = x_0(t) + x_p(t) = 2C_1 e^t \cos 2t + 2C_2 e^t \sin 2t + 2e^t \\ y(t) = y_0(t) + y_p(t) = C_1 e^t(\cos 2t - \sin 2t) + C_2 e^t(\cos 2t + \sin 2t) + e^t \end{cases}$

$$\begin{cases} x(0) = 0 \\ y(0) = 0 \end{cases} \Rightarrow \begin{matrix} 2C_1 + 2 = 0 \\ C_1 + C_2 + 1 = 0 \end{matrix} \Rightarrow \begin{matrix} C_1 = -1 \\ C_2 = 0 \end{matrix}$$

$\Rightarrow \begin{cases} x(t) = \dots \\ y(t) = \dots \end{cases}$

Variația $\begin{cases} x_0(t) = 2C_1 e^t \cos 2t + \dots \\ y_0(t) = C_1 e^t(\cos 2t - \sin 2t) + \dots \end{cases}$

Caut $\begin{matrix} x_p(t) \\ y_p(t) \end{matrix}$ de aceeași formă cu termenul liber $\begin{pmatrix} 0 \\ 2e^t \end{pmatrix}$

$x_p(t) = ae^t$ $\quad = 2e^t$ $\qquad y_p(t) = be^t$ $\quad = e^t$ $\qquad \lambda = 1$ nu e val. pr. pt. $\begin{pmatrix} -1 & 4 \\ -2 & 3 \end{pmatrix}$

$$\begin{cases} x' = -x + 4y \\ y' = -2x + 3y + 2e^t \end{cases} \qquad \begin{cases} a = -a + 4b \\ b = -2a + 3b + 2 \end{cases}$$

$2a = 4b \quad a = 2b$

$-2b + 2a = 2$

$-2b + 4b = 2$

$b = 1$

$a = 2$

Atenție: ac. variantă nu merge aceea:

$$\begin{cases} x' = x + 2y + 3e^{-t} \\ y' = 2x + y - e^{-t} \end{cases}$$

$x_p(t) = ae^{-t}$

$y_p(t) = be^{-t}$

$\lambda = -1$ val. pr. pt matrice ext. $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

$$\begin{cases} -ae^{-t} = ae^{-t} + 2be^{-t} + 3e^{-t} \\ -be^{-t} = 2ae^{-t} + be^{-t} - e^{-t} \end{cases}$$

$$\begin{cases} 0 = 2a + 2b + 3 \\ 0 = 2a + 2b - 1 \end{cases}$$

sistem incompatibil